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Liquidity shocks: A new solution to the forward premium puzzle☆

Vikram Kumar

Davidson College, Box 6975, Davidson, NC, 28035-6975, USA

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ABSTRACT

The frequent empirical failure of uncovered interest rate parity raises a question that has not been definitively answered: why do predictable excess returns on currencies persist in competitive currency markets? Supported by data from nine major currencies for 1978:08–2019:09, I provide a novel resolution to this enduring forward premium puzzle by building on the financial economics literature that explores the economic implications of limited access to capital markets. A liquidity shock, or the urgent demand for liquidity by credit-constrained arbitragers liquidating bond holdings, causes losses from sudden drops in bond prices. Arbitragers require a liquidity premium to compensate for potential losses that vary directly with the interest rate. It is this liquidity premium that explains persistent excess returns on currencies. I argue for policies favoring a low interest rate environment and macroprudential controls that ease liquidity constraints to increase the efficiency of international capital markets by reducing the liquidity premium.

1. Introduction

In markets with liquidity constraints, liquidity shocks that necessitate asset liquidations can result in sudden drops in asset prices. In this scenario, those investors who are forced to liquidate their default-free bonds immediately would obtain a price below the face value. Therefore, instead of the returns they would get if they held the bond to maturity, they would stand to make losses were a liquidity shock to occur. The potential for such losses changes the incentives for international arbitrage because expected returns on bonds also incorporate investors' expectations about the liquidity shock.

This paper examines the impact of liquidity shocks on the uncovered interest parity (UIP) condition which guides international capital flows. The UIP condition says that the expected change in the exchange rate equals the interest rate differential between risk-free domestic and foreign bonds. In other words, the high interest rate currency should be expected to depreciate to the point that the same-currency returns on high and low interest rate bonds become equal. But empirical tests of UIP very often find that high interest rate currencies do not depreciate sufficiently, or even appreciate *ex post*. This implies the existence of predictable positive excess returns on the high interest rate currency, and negative excess returns on the low interest rate currency. In regressions of one-period ahead changes in the exchange rate on the interest rate differential, UIP predicts a slope coefficient of one. However, it is well-documented in the literature that, empirically, the slope coefficient is often less than unity, and frequently negative. This persistent downward bias in the slope coefficient is called the forward premium puzzle.

In this paper I show that anticipated liquidity shocks cause the UIP condition to fail, as in the data. What is the intuition behind this? If there is no liquidity shock, then zero-coupon bonds are held to maturity and earn a known interest rate. If a liquidity crisis occurs, the yield is negative because, in liquidity-constrained markets, the fire-sale liquidation of securities prior to maturity causes losses from sudden declines in asset prices. So, the *expected* return is less than the interest rate for both home and foreign bonds. I assume that losses in the bad state are directly proportional to interest rates in the good state, or that 'the higher they are, the harder they fall.' Zero-arbitrage occurs when the *expected* return on domestic bonds equals the *expected* return on foreign bonds plus the expected appreciation of the foreign currency.

To illustrate why the slope coefficient will be biased, let us say the home (foreign) interest rate is 10% (6%). In this case, UIP will predict that, on average, the foreign currency will appreciate by 4% [= 10%–

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E-mail address: vikumar@davidson.edu.

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6%]. However, a risk-neutral investor will base her decision on *expected* yields. If the corresponding *expected* yield on home (foreign) currency bonds is 8% (4.8%), then the data, on average, will show that the foreign currency appreciates only by 3.2% [= 8%–4.8%]. So, on average, the data will show a predictable positive excess return on the high-interest home currency of 0.8% [= 10%-(6% + 3.2%)]. This example is consistent with a UIP regression slope coefficient of 0.8. The deviation from unity of −0.2% is a liquidation premium, required to offset the larger loss in the crisis for the investor holding the high interest asset. Further, if the investor is risk-averse and places a sufficiently high utility value on the loss from the liquidity shock, then she will hold the low interest asset even if that currency is expected to depreciate because it will carry a high liquidity discount. In this case the slope coefficient will be negative.

To characterize the arbitragers' payoff in a liquidity crisis, I utilize the survey-with-a-model framework of Krishnamurthy (2010), who identifies the key determinants of sudden declines in asset prices due to adverse liquidity events. More broadly, by presuming that arbitragers can be credit-constrained in some states of the world, I build on the financial economics literature that explores the economic implications of limited access to internal capital markets, such as Eisfeldt and Rampini (2007); Brunnermeier et al. (2008), who examine currency crash risk when speculators near funding constraints; and Caballero and Krishnamurthy (2003), who show that collateral constraints affect investors' relative demand for domestic currency and dollar debt.

These findings complement other explanations of the forward-premium puzzle that typically assume either rational expectations and time-varying risk premium (Backus et al. (2001); Verdelhan (2010)), or expectations errors (Bacchetta and Van Wincoop (2010), Evans and Lewis (1995)), which allow excess returns to persist. The former requires risk-aversion, the latter does not. In suggesting a new channel for the failure of UIP, I show that currency forecast errors will contain systematic variations in the liquidity premium, so the forward premium cannot be a rational forecast of the currency depreciation. Beyond that, the slope coefficient will be less than unity but positive under risk neutrality; it can be negative only under risk-aversion. In the former case, risk-neutral investors require a liquidation premium to compensate for the potential crisis-state outcome (see Caballero and Krishnamurthy (2003)); in the latter case, risk-averse investors seek an additional liquidity risk premium. In this sense, my model integrates both types of explanations (see section 4).

Though evidence over the past few decades, starting with Hansen and Hodrick (1980), Fama (1984) and others, has cast doubt on the empirical validity of UIP, this is not a universal finding. For instance, Boudoukh et al. (2005) and Chinn (2006) find that UIP tends to hold up well for long horizons; Lee (2011) fails to find systematic deviations from UIP in a heterogeneous sample of currencies; and Cover and Mallick (2012) do not find support for the failure of UIP in UK data. Similarly, Jiang et al. (2013) and Bhatti (2014) fail to find evidence of deviations from UIP for several Eastern European currencies, and Hassan and Mano (2019) are unable to reject UIP upon correcting for uncertainty regarding future mean interest rates. Yet, the documented failure of UIP remains a key price puzzle in international economics and finance. This paper contributes to the literature that studies the economic implications of limited internal capital markets, in this case the arbitragers' limited capital during times of crisis.

In section 2, the framework in Krishnamurthy (2010) is used to model the arbitragers' payoffs in the bad state when a liquidity shock occurs. In section 3, the investors' zero-arbitrage conditions are determined and a liquidity risk-augmented UIP condition is derived. Its relationship with the forward premium puzzle is examined in section 4. In section 5, 41 years of data on nine major currencies are used to assess the results in the context of the forward premium puzzle, and policy suggestions are provided. Concluding remarks and directions for further research are in section 6.

2. Interest arbitrage with liquidity risk

2.1. Home investor & liquidity shock

A Home investor can buy a Home country security and a Foreign country security with identical default risk. Liquidity conditions in Home and Foreign markets are symmetric. The investor confronts two possible states of the world *ex ante*. No liquidity shock occurs in the good state (G) and the investor carries the bond to maturity, redeeming it at face value. This state unfolds with probability $(1-\pi)$. In the bad state (B), which occurs with probability π, an exogenous liquidity shock compels the investor to sell the asset at a discount. Denote as R^G and R^B the asset returns conditional on the realization of states G and B, respectively. I hypothesize that R^G and $R^B$are inversely proportional, or that 'the higher they are, the harder they fall.' Consider below the Home investor's problem.

In the good state of the world there is no liquidity shock. The Home security is purchased in period t at price P_t, and redeemed at maturity in period $t+1$ at face value F. Normalize prices such that $F = 1$. Then the return in the good state is simply the nominal interest rate $r_t = (1/P_t) - 1$. Because $(1+r_t) = (1/P_t)$, we can write $\ln(1+r_t) = -\ln P_t$. Further, since $\ln(1+r_t) \approx r_t$ for small r_t, we obtain $r_t = -\ln P_t$. Note that $P_t < 1$ so $\ln P_t < 0$. The rate of return on the asset in the good state is

$$R_t^G = r_t = -\ln P_t \tag{1}$$

In the bad state, liquidity conditions in the market determine the asset price $P_{t+\tau}$ at the time of sale $t+\tau$ prior to $t+1$, where $0 < \tau < 1$. The cash from the liquidation simply funds the liquidity contingency. In what follows, I assume that the investor knows the parameters of the bad state and so knows $P_{t+\tau}$, but does not know if the bad state will be realized. The return conditional on the bad state is. $R_t^B = \ln P_{t+\tau} - \ln P_t$.

The key issue is to obtain the value of $P_{t+\tau}$, and I utilize the survey-with-a-model framework in Krishnamurthy (2010) to do so. Assume that a unit mass of hedge funds are the investors. They purchase bonds in period t at price P_t and raise funds through equity and debt to buy one unit of the asset which may or may not be required to be liquidated. Assume that immediately upon purchasing the asset, φ percent of hedge funds receive exogenous liquidity shocks, causing them to sell their holdings of the asset. In this sense, φ is one element of the size of the liquidity shock in terms of the proportion of assets impacted. The shock is distributed randomly among hedge funds so they do not know if they will, or will not, be forced to liquidate. Assume that the $(1-\varphi)$ percent of the hedge funds that are not shocked do not liquidate any asset.[1] Total assets on offer for sale, or the supply of assets, at $t+\tau$ is

$$L_{t+\tau}^S = \varphi \tag{2}$$

There are willing deep-pocket buyers of these assets whose own funds are in limited supply. We assume that (a) their supply of liquidity to absorb assets on offer is not infinitely elastic, and (b) the loss of hedge funds in the bad state is proportional to asset returns in the good state. To model these considerations, assume that the deep-pocket buyers' inverse demand function for assets is

$$P_{t+\tau} = P_t^{1+\delta L_{t+\tau}^D} \tag{3}$$

In (3) the parameter $\delta > 0$. Recall that the security has a face value of 1, and $P_t < 1$. Since the exponent on the RHS exceeds unity, deep-pocket investors will provide liquidity in the amount $L_{t+\tau}$ only if the price falls below P_t, i.e., $P_{t+\tau} < P_t$. This fall in price is larger the larger is δ, so in this sense δ is a second element determining the size of the liquidity shock in

[1] Equations and in this study algebraically depict Fig. 1 in Krishnamurthy (2010) excluding endogenous liquidation elements of that figure.

addition to φ.

Setting $L^S_{t+\tau} = L^D_{t+\tau}$, by equations (2) and (3) the price of the security in period $t+\tau$ is given by

$$P_{t+\tau} = P_t^{1+\delta\varphi} \tag{4}$$

In (4), the effect on the asset price at time $t+\tau$ depends on the term $\delta\varphi$. Let us define the *size of the liquidity shock* as

$$\ell \equiv \delta\varphi \tag{5}$$

Therefore, (4) implies that $\ln P_{t+\tau} = (1+\ell)\ln P_t$, and the return in the bad state is $R^B_t = \ln P_{t+\tau} - \ln P_t = \ell \ln P_t$. Since $r_t = -\ln P_t$ by (1) we obtain

$$R^B_t = \ln P_{t+\tau} - \ln P_t = -\ell\, r_t \tag{6}$$

Equations (1) and (6) tell us that the loss in the bad state is larger if the return in the good state is larger.

Now consider a Foreign security held by the Home investor. This investor purchases foreign currency at the spot exchange rate S_t (the Home currency price of Foreign currency), and then the security at the Foreign currency price P^*_t (asterisk denotes foreign variables). For reasons noted above, the foreign interest rate is $r^*_t = -\ln P^*_t$. With tilde denoting random variables, the depreciation rate of the Home currency conditional on the information at t is

$$\tilde{d}_{t+1} = \tilde{S}_{t+1} S_t^{-1} - 1 \tag{7}$$

In the good state, the return on the Foreign security, not known to the Home investor *ex ante*, is

$$\tilde{R}^G_t = r^*_t + \tilde{d}_{t+1} \tag{8}$$

Liquidity conditions in Home and Foreign markets are assumed to be symmetric.[2] In the bad state, the φ percent of the firms subject to liquidity shocks sell assets at the foreign currency price $P^*_{t+\tau}$, and then convert the resulting foreign exchange at the unknown spot rate $\tilde{S}_{t+\tau}$. Though the investor in the bad state is subject to both a liquidity risk and a foreign exchange risk, I focus the analysis solely on liquidity risk. I do so by assuming that currency markets are not liquidity-constrained, and $\tau \to 0$, i.e., the liquidity shock, were it to occur, occurs with the smallest possible lapse of time after the asset purchase so that proximately$S_{t+\tau} = S_t$.[3] So the hedge fund converts foreign currency holdings of $P^*_{t+\tau}$ into domestic currency at the known spot rate S_t, avoids exchange risk altogether, and faces only liquidity risk. With this simplification, the investor's capital gain is $\ln P^*_{t+\tau} - \ln P^*_t = \ell \ln P^*_t$, and since $r^*_t = -\ln P^*_t$, we obtain

$$R^{*B}_t = \ln P^*_{t+\tau} - \ln P^*_t = -\ell\, r^*_t \tag{9}$$

Let the investor's preference conditional on the state of the world be given by $u(y)$, where y is the relevant return on the asset, $u(0) = 0$, and $u'(y) > 0$. Noting that π is the probability of a liquidity shock, by (1) and (6), the Home investor's unconditional expected utility from the Home security (H) is

$$E\left(u^H_t\right) = (1-\pi)u(r_t) + \pi u(-\ell\, r_t) \tag{10}$$

Regarding the Home investor's holding of the Foreign asset (F), per (8), the expected utility conditional on the good state is $E(u^F_t|G) = E(u(r^*_t + \tilde{d}_{t+1}))$. In certainty equivalent terms, $E(u^F_t|G) = u(r^*_t + E(\tilde{d}_{t+1}) - \mu_t)$where μ_t is the Arrow-Pratt risk premium required to avoid exchange risk when the Foreign security is redeemed at maturity. Note that $\mu_t = \left(1/2\right)\lambda V(\tilde{d}_{t+1}|I_t)$, where $\lambda = -u''(\cdot)/u'(\cdot)$ is the risk aversion parameter, and $V(\tilde{d}_{t+1}|I_t)$ is the conditional variance of exchange rate depreciation given the information at t.

Further, per (9), the expected utility conditional on the bad state, when a liquidity shock occurs, is $E(u^F_t|B) = u(-\ell\, r^*_t)$. Thus, the Home investor's unconditional expected utility from the Foreign (F) asset is

$$E\left(u^F_t\right) = (1-\pi)u\left(r^*_t + E(\tilde{d}_{t+1}) - \mu_t\right) + \pi u\left(-\ell\, r^*_t\right) \tag{11}$$

2.2. Foreign investor & liquidity shock

I now turn to the problem of the Foreign country hedge fund. First consider the returns from its holding of own-currency asset in the good state, when there is no liquidity shock. Having bought the Foreign asset at price $P^*_t < 1$, the Foreign investor redeems the asset at face value $F = 1$. No foreign exchange transaction is involved. Consequently, the rate of return is $r^*_t = (1/P^*_t) - 1$. Because $\ln(1 + r^*_t) = -\ln P^*_t$, and $\ln(1+r^*_t) \approx r^*_t$ for small r^*_t, the good state rate of return to the Foreign investor on the Foreign security is as given in (12), where the caret denotes the Foreign *investor's* variables.

$$\hat{R}^{*G}_t = r^*_t = -\ln P^*_t \tag{12}$$

Note that $\ln P^*_t < 0$ since $P^*_t < 1$. By symmetry with equation (9), in the bad state, the return is

$$\hat{R}^{*B}_t = -\ell\, r^*_t \tag{13}$$

The Foreign investor's return on the Home asset in the good state must take account of the depreciation of the Home currency since it is held to maturity, and it equals $r_t - \tilde{d}_{t+1}$. If the bad state unfolds, prompting liquidation by φ percent of hedge funds, then the return equals $-\ell\, r_t$. Thus

$$\hat{\tilde{R}}^G_t = r_t - \tilde{d}_{t+1} \tag{14}$$

$$\hat{R}^B_t = -\ell\, r_t \tag{15}$$

For reasons articulated in the previous section, with identical preferences of Home and Foreign investors, (12)–(15) yield the unconditional expected utility from the Foreign investor's holdings of the Foreign asset as

$$E\left(\hat{u}^F_t\right) = (1-\pi)u\left(r^*_t\right) + \pi u\left(-\ell\, r^*_t\right) \tag{16}$$

and from the Home asset as

$$E\left(\hat{u}^H_t\right) = (1-\pi)u(r_t - E(\tilde{d}_{t+1}) - \mu_t) + \pi u(-\ell\, r_t) \tag{17}$$

3. Zero-arbitrage conditions & financial market equilibrium

The investor's equilibrium requires equality of expected utilities from holdings of Home and Foreign assets. Using (10)-(11) the zero arbitrage condition for the Home investor holds when $E(u^H(r_t)) - E(u^F(r^*_t)) = 0$. Likewise, using (16)-(17) the Foreign investor's zero arbitrage condition is$E(\hat{u}^F(r^*_t)) - E(\hat{u}^H(r_t)) = 0$.[4] Note that investors will participate in the market only if $E(u^H(r_t)) > u(0)$ in (10), and $E(\hat{u}^F(r^*_t)) > u(0)$ in (16). The

[2] Given significant financial market integration, and following Verdelhan (2010), I assume Home and Foreign structural parameters to be the same throughout this paper.

[3] This assumption enables me to derive a closed-form expression for a liquidity risk-augmented UIP condition identifying a mechanism for deviations from UIP that is distinct from those in the literature such as in Fama (1984), Evans and Lewis (1995), Backus et al. (2001) and Bacchetta and Van Wincoop (2010).

[4] The caret refers to the Foreign *investor* and * refers to the Foreign *asset*.

necessary and sufficient conditions for these inequalities to hold are derived in Appendix A.

In financial market equilibrium, Home and Foreign investors will *both* earn zero excess returns simultaneously in expected utility terms. Derived in Appendix B, the international financial market equilibrium condition is

$$E(\tilde{d}_{t+1}) = \left[1 - \frac{\pi \ell}{1-\pi}\overline{\theta}(r_t, r_t^*)\right](r_t - r_t^*)$$
$$\overline{\theta} = \frac{1}{2}\left[\theta(r_t) + \theta(r_t^*)\right] \tag{18}$$

In the equation above $\theta(\cdot)$ is the investors' marginal utility from own-country asset returns when there is a liquidity shock relative to when there is no liquidity shock. It is a pseudo-MRS term since a marginal rate of substitution in form but without choice variables. Specifically:

$$\theta(y_t) = \frac{u'(-\ell y_t)}{u'(y_t)} \geq 1 \tag{19}$$

Under risk-neutrality the risk aversion parameter $\theta(\cdot) = 1$.[5] Under risk-aversion $\theta(\cdot) > 1$, reflecting the investor's subjective valuation of utility loss at the margin if a liquidity shock occurs relative to utility gains at the margin if there is no liquidity shock. With $\theta'(\cdot) > 0$, the relative utility value of the loss *increases with the interest rate level.* This analysis thus highlights the property that the desirability of holding the high interest asset is tempered by the higher loss it brings if a liquidity crisis occurs.

4. Liquidity risk-augmented uncovered interest parity $\ell -$ UIP

When there is no liquidity risk, $\pi = 0$ and (18) reduces to the canonical UIP condition $E(\tilde{d}_{t+1}) = r_t - r_t^*$. However, with liquidity risk (18) becomes the liquidity risk-augmented UIP (or $\ell -$ UIP) condition.

4.1. Liquidity premium

The $\ell -$ UIP condition is symmetric from the perspectives of both Home and Foreign investors. It shows that the expected currency depreciation rate equals the interest rate differential *adjusted for a liquidity premium.* Let us write (18) in terms of excess returns as follows:

$$E(\tilde{d}_{t+1}) = (1 + \beta_t)\left[r_t - r_t^*\right] \tag{20}$$

where

$$\beta_t = -\frac{\pi \ell}{1-\pi}\overline{\theta}(r_t, r_t^*) < 0 \tag{21}$$

Because β is negative, we see in (20) that the slope coefficient on the interest rate differential is less than one, and this is so under both risk neutrality and risk aversion.[6] But under risk-aversion $\overline{\theta}$ can be large enough to make the slope coefficient negative. Specifically, we see that deviations from UIP are caused by a time-varying liquidity premium given by

$$\textit{liquidity premium}_t = \beta_t(r_t - r_t^*) \tag{22}$$

The size and sign of the liquidity premium depend upon the size and sign of the interest rate differential. Under risk aversion, its size also varies with interest rate *levels* because β varies with interest rate levels. In utility terms at the margin, the higher interest rate asset produces a larger loss $-\ell r$ or $-\ell r^*$ in a liquidity crisis per (6) and (9).[7] Therefore, investors require a larger liquidity premium on the higher interest asset to compensate them for its higher perceived loss in terms of relative utility value $\overline{\theta}$.

Since β is negative, if the Home interest rate is higher than the Foreign rate, then $r_t > r_t^* + E(\tilde{d}_{t+1})$: the expected depreciation of the Home currency will not fully offset the interest rate differential due to the required liquidity premium on Home bond. Thus, the liquidity premium will show up in the data as excess returns on Home bonds. The Home currency may even be expected to appreciate when β is sufficiently negative. This happens when the loss during the liquidity crisis, valued in utility terms, is sufficiently high.

4.2. Relationship with forward premium puzzle

The $\ell -$ UIP condition incorporates a new channel for the failure of UIP and provides a new framework for understanding how the forward premium puzzle can arise from anticipated liquidity shocks. Because liquidity shocks change the *expected* returns from default-free bonds, they induce the investor to seek a liquidity premium. This liquidity premium manifests itself in the form of persistent excess returns on currencies, or deviations from UIP. Empirical deviations from UIP are the focus of the large literature on the forward premium puzzle (e.g., Fama (1984)) which postulates that, under rational expectations and risk aversion, a time-varying risk premium explains deviations from UIP. However, modelling the risk premium empirically has proved difficult.[8] In $\ell -$ UIP, the liquidity premium $\beta(r - r^*)$, embodies deviations from UIP.

Because covered interest parity is a virtual empirical identity, define the forward premium as $\ln X_t - \ln S_t \equiv x_t = r_t - r_t^*$, where X_t is the one-period-ahead forward rate. Then (20) may be re-written as

$$E(\tilde{d}_{t+1}) = (1 + \beta_t)x_t \tag{23}$$

Conventional UIP regressions of d_{t+1} on x_t assume that the slope coefficient of x_t is one. Under rational expectations and risk neutrality this implies that the forward premium is an efficient forecast of spot rate depreciation. Yet (23) shows that under $\ell -$ UIP, the slope coefficient $(1 + \beta_t)$ is biased below one since $\beta_t < 0$, so β_t is a measure of the *forward premium bias*. In (23) the liquidity premium $\beta_t x_t$ is time-varying in size and sign, and it is negatively correlated with the forward premium x_t given that $\beta_t < 0$. This reflects well-documented properties of Fama's risk premium[9] and suggests a channel for deviations from UIP that is new in the literature.

Thus, under $\ell -$ UIP x_t cannot be a rational forecast of d_{t+1} because forecast errors will contain the liquidity premium $\beta_t x_t$ and be systemat-

[5] $\theta'(y_t) = \lambda(1+\ell)\theta(y_t)\ \forall y_t = r_t, r_t^*$. $\lambda = \theta'(y_t) = 0$ under risk neutrality. Under risk aversion, $\lambda > 0$ and $\theta' > 0$ so $\theta > 1$. See Appendix C for derivation of θ'.

[6] Under risk neutrality $\overline{\theta} = 1$ and due to the sufficiency condition in Appendix A, $0 > \beta_t > -1$. But under risk-aversion $\overline{\theta}$ can be large and it is possible that $\beta_t < -1$.

[7] Recall that $\ell = \delta\varphi$, indicating that the loss is larger the larger the proportion φ of hedge funds receiving the liquidity shock, and the stronger is the liquidity constraint reflected in a larger value of δ.

[8] Models of risk premium include Backus et al. (2001), who examine the role of the term structure of interest rates in the failure of UIP; Lustig and Verdelhan (2007), who show that excess returns offset the cost of consumption growth risk faced by investors; Verdelhan (2010), who examines the role of habit and varying risk-aversion in the presence of consumption growth risk; and Lustig and Verdelhan (2019), who examine the effect of incomplete spanning on the currency risk premium. In this vein Nagayasu (2014) has noted that "the risk premium seems to be closely associated with liquidity constraints," and Engel et al. (2019) have argued that excess returns on foreign bonds vary directly with the US and foreign interest rate differentials due to the role of 'liquidity returns' on the dollar.

[9] The definition of the risk premium in Fama is $x_t - d_{t+1}$, which is of the opposite sign relative to this paper (Fama, 1984, 320).

ically time-varying.[10] Importantly, forecast errors will not be white-noise *even under risk-neutrality* since β_t is always negative.[11] Further, the slope coefficient of x_t is *less than one regardless of risk preferences*, though it cannot be negative *without* risk-aversion because only then can $\beta_t < -1$ as shown above. Negative estimates of the slope coefficient have been common as reported in Table 1 below. Under $\ell-$UIP, a key variable influencing the slope coefficient $(1+\beta)$ is the preference curvature term $\overline{\theta}$. If investors perceive a tail risk of a large liquidity shock ℓ and interest rate levels are high enough, then the value of $\overline{\theta}$ can be large such that $\beta < -1$.

5. Empirical support & policy implications

5.1. Empirical support

In this section, I use data for nine major currencies to illustrate and estimate the slope coefficient of the forward premium in UIP regressions. I contextualize the results in other findings in the literature, and then assess how well the implied slope coefficient under the $\ell-$ UIP framework compares with those estimates. I use data for 1978:08–2019:09 for the Belgian Franc (BEF), Canadian Dollar (CAD), Swiss Franc (CHF), Deutschemark (DEM), French Franc (FFR), British Pound (GBP), Italian Lira (ITL), Japanese Yen (JPY) and Dutch Guilder (NEG). Spot rates are US Dollar (USD) prices of these currencies, and interest rates are monthly one-month Eurocurrency deposit rates (see Data Appendix for details).

Fig. 1 presents the relationship between *ex-post* realizations of one-period-ahead currency depreciation rates $\overline{d}_{+1}$ (scaled by 100), and the forward premium given by the interest rate differentials $\overline{r}^{USD}-\overline{r}^*$ (scaled by 100). UIP predicts that, on average, the observations will lie along the 45°-line if on average $\overline{d}_{+1}$ reflects $E(\tilde{d}_{+1})$. Systematic deviations from the 45°-line suggest the failure of UIP. These data show that for all currencies except the French Franc, $\overline{d}_{+1} < (\overline{r}^{USD}-\overline{r}^*)$ when $(\overline{r}^{USD}-\overline{r}^*) > 0$, and $\overline{d}_{+1} > (\overline{r}^{USD}-\overline{r}^*)$ when $(\overline{r}^{USD}-\overline{r}^*) < 0$. Thus, the average *ex-post* currency depreciation for eight of the nine currencies was less than the average interest rate differential. This indicates support for the $\ell-$ UIP theory that the slope coefficient of the UIP regression is less than one. Within the currency sample, the Lira and the Pound lie closest to the 45°-line, and the Yen and Swiss Franc the farthest.

For the currency sample in Fig. 1, the first row of Table 1 reports the range of estimated slope coefficients $(1+\widehat{\beta})$ from UIP regressions using the estimating equation for (20).[12] The point estimates of slope coefficients for these nine currencies range from -0.97 (Yen) to $+0.61$ (Lira) which seem consistent with the data in Fig. 1. The restricted estimate of the slope coefficient is 0.13. Other rows of Table 1 show estimates of slope coefficients from selected studies in the forward premium puzzle literature. For different currency samples and sample periods the estimates range from -0.21 to -3.50. Overall, all estimates are less than one, and most are negative.

A gauge of strength of the $\ell-$UIP model is the extent to which the empirical estimates in Table 1 are reflected in the model's implied values of the slope coefficient for reasonable parameter values in a parsimonious setting. Assume that investors have exponential utility: $u(y) = \lambda^{-1}(1-\exp(-\lambda y))$. Then (19) implies that $\overline{\theta} = \exp(\lambda(1+\ell)y)$ for $y = r = r^*$. Assume that Home and Foreign investors have identical preferences and they perceive a tail risk of a liquidity shock of size ℓ. To first fix ℓ, let us assume that the liquidity crisis occurs when asset prices, or interest rates, are at a normal level. Taking the median US interest rate of 5% over the sample period 1978:08–2019:09 as a good measure of 'normal,' the initial bond price is $P_t = 0.952$. Now assume that a liquidity shock of size ℓ results in a k% reduction in this bond price ($k = 5\%, 10\%, \ldots, 95\%$). Then it can be shown that[13]

$$\ell = \frac{\ln(1-k)}{\ln(0.952)} \tag{24}$$

Each value k in (24) yields an associated value of ℓ which is then held constant. Now, for each given liquidity shock of size ℓ, the corresponding value of the slope coefficient $(1+\beta)$ is obtained using (21). Values of $(1+\beta)$ are obtained assuming that the tail risk is embodied in low liquidity shock probabilities $\pi = (1\%, 2\%)$; for risk aversion values $\lambda = (0, 2, 5)$; and interest rates $r = (3\%, 5\%, 10\%)$. Recall that interest rate levels affect $(1+\beta)$ via $\overline{\theta}$.

The results are displayed in Table 2. Blank cells indicate that the investor is better off holding a cash position and not participating in the market.[14] These results correspond closely with the empirical estimates of the slope coefficient in Table 1. They show that tail risks of large losses during liquidity crises result in deviations from UIP like those in the currency sample in this study and others reported in the literature. Under risk neutrality $1 > (1+\beta) > 0$: the slope coefficient is positive but less than one, and it becomes smaller the larger is the liquidity shock ℓ. Under risk aversion, the forward premium bias β in absolute value increases with interest rate levels reflecting 'the higher they are, the harder they fall' analogy. And some combinations of increases in the size and probability of the liquidity shock, investor risk aversion, and interest rate levels conspire to make the slope coefficient $(1+\beta) < 0$, indicating that the higher-interest currency is expected to appreciate.

5.2. Policy implications

These results have useful policy implications for reducing the forward premium bias or deviations from UIP, which is important for achieving efficiency in the international allocation of capital. Our analysis of liquidity factors has shown that the required liquidity premium interferes with the operation of UIP. The presence of a liquidity premium reduces the space for exchange rates to adjust fully to interest rate differentials shown by (20) & (22). This, in turn, means that the international flow of funds is sub-optimal, with funds locked in less efficient uses due to the liquidity premium wedge in UIP. The $\ell-$UIP framework offers guidance for policymakers about approaches to reducing the liquidity premium and thereby increasing financial market efficiency.

Firstly, the results show that if interest rates are low, then deviations from UIP are also low. As discussed above, this beneficial outcome recommends a low interest rate policy environment. Secondly, deviations from UIP can be reduced if policymakers have in place effective macroprudential controls. Specifically, ensuring continued supplies of liquidity

[10] For instance, the existence of time-varying coefficients has been documented by Bansal (1997), Baillie and Bollerslev (2000), and Baillie and Killiç (2006), who find among them notable differences in the patterns of deviations from UIP depending upon the sign and absolute size of x_t, and regime transitions based among other things on differentials in incomes and in monetary growth rates. However, none posit conditioning the coefficient on interest rate levels as this paper does.

[11] The role of systematic forecast errors is explored in Evans and Lewis (1995), who argue that with expectations of regime-switching in the long run, small sample serial correlation in forecast errors can produce evidence of persistent excess returns on currencies, and Bacchetta and Van Wincoop (2010), who argue that infrequent portfolio adjustments due to adjustment costs lead to predictable excess returns on currencies. Froot and Frankel (1989) fail to find the existence of risk premia in survey data, and Burnside et al. (2011) find that excess returns are unrelated to traditional risk factors.

[12] The reported estimates are for $d_{t+1} - (r_t^{USD} - r_t^*) = \alpha + \beta(r_t^{USD} - r_t^*) + \varepsilon_{t+1}$ using Seemingly Unrelated Regressions with both restricted and unrestricted β and contemporaneous covariances. The slope coefficient reported in Table 1 is $(1+\widehat{\beta})$. See notes to Table 1 and Data Appendix for details.

[13] In terms of, $P_{t+\tau} = (1-k)P_t$, or that $\ln P_{t+\tau} - \ln P_t = \ln(1-k)$. But because $r_t = -\ln P_t$, (6) then implies that $\ln(1-k) = \ell \ln P_t$ which yields (24).

[14] See discussion in section 3. Necessary and sufficient conditions are in Appendix A.

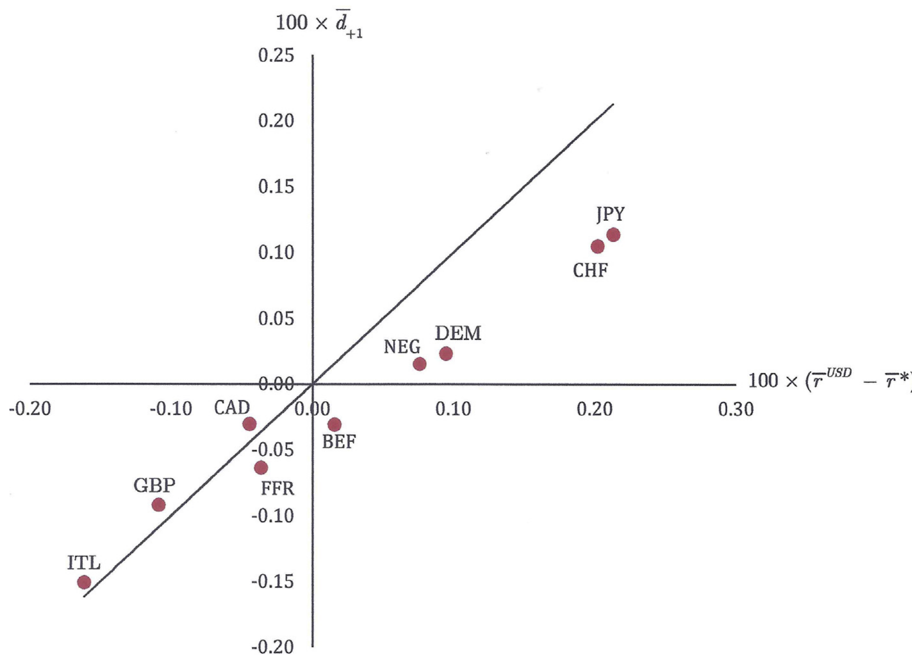

Fig. 1. Mean One-Period Ahead Currency Depreciation Rate v. Mean Interest Rate Differential for Nine Currencies, 1978:08–2019:09
†Notes:

- $\overline{d}_{+1}$ is the sample mean, scaled by 100, of the one-period ahead monthly percentage change in spot rates (USD per foreign currency) of the Belgian Franc (BEF), Canadian Dollar (CAD), Swiss Franc (CHF), Deutschemark (DEM), French Franc (FFR), British Pound (GBP), Italian Lira (ITL), Japanese Yen (JPY) and Dutch Guilder (NEG).
- $(\overline{r}^{USD} - \overline{r}^{*})$is the sample mean forward premium, or the interest rate differential, scaled by 100, between the monthly one-month Eurocurrency deposit rates on USD and the foreign currency deposits.
- $\overline{d}_{+1} = (\overline{r}^{USD} - \overline{r}^{*})$along the solid line. For each sample currency over this sample period except the French Franc, the mean currency depreciation (appreciation) was less than the corresponding mean interest rate differential. The discrepancy was the smallest for CAD, ITL and GBP and the largest for CHF and JPY.

Table 1
Ranges of estimated slope coefficient $(1+\beta)$ of the forward premium in UIP regressions: Selected studies††

Study	Table Reference	Period	Currencies	Range of Estimated Slope Coefficients	
				Lowest	Highest
This paper.[a]		1978:08–2019:09	BEF, CAD, CHF, DEM, FFR, GBP, ITL, JPY, NEG	0.61***	−0.97***
				Restricted SUR: 0.13***	
Bachetta, P. and van Wincoop, D. (2010)	Table 1	1978:12–2005:12	CAD, CHF DEM, GBP, JPY	−0.55***	−3.06***
Backus, D. et al. (2001)	Table 1	1974:07–1994:11	DEM, GBP, JPY	−0.74	−1.84***
Evans M., and Lewis, K. (1995)	Table 4	1975:01–1989:01	DEM, GBP, JPY	−2.02***	−3.50***
Fama, E.F. (1984)	Table 4	1973:08–1982:12	BEF, CAD, CHF, DEM, FFR, GBP, ITL, JPY, NEG	−0.21	−1.15**
				Restricted SUR: -0.58***	

††Notes: *** and ** mean 1% and 5% level of significance. The original UIP estimating equation is either of the form $d_{t+1} = \alpha + (1+\beta)(r_t - r_t^*) + \gamma_{t+1}$ with $H_0 : (1+\beta) = 1$, or of the form $d_{t+1} - (r_t - r_t^*) = \alpha + \beta(r_t - r_t^*) + \varepsilon_{t+1}$ with $H_0 : \beta = 0$. In either case $(1+\widehat{\beta})$ is shown above. SUR means Seemingly Unrelated Regression. Restricted means β is restricted. d_{t+1} is the 1-month percent change in the (USD/foreign currency) spot rate. In all cases the estimates of the slope coefficient $(1+\beta) < 1$. Since $\beta < 0$, it is shown in section 4.2 that the liquidity premium βx and forward premium $x = (r - r^*)$ are negatively correlated.

[a] This paper: Seemingly Unrelated Regression with contemporaneous covariances between currencies obtained using the estimating equation $d_{t+1}^j - (r_t^{USD} - r_t^j) = \alpha^j + \beta^j(r_t^{USD} - r_t^j) + \varepsilon_{t+1}^j$. $\overline{R}^2 = 0.11$ and 0.12 for restricted & unrestricted regressions, respectively. Constants (not reported) are not restricted and are not statistically different from zero in all cases. Robust panel corrected standard errors utilized. Estimates corrected for serial autocorrelation. Panel unit root tests (not reported) reject presence of unit roots assuming common and individual unit roots in the sample. Number of observations in the pooled sample is 4428 after adjustment. Currency codes are described in section 5. Additional details are in the Data Appendix.

by maintaining liquidity buffers, and releasing them under adverse shocks, will limit the size of the liquidity shock ℓ. Note also that easing liquidity constraints can prevent the slow build-up over time of enabling conditions for financial market instability and currency crashes as discussed by Brunnermeier et al. (2008). Thirdly, findings by Jiménez-Martín and Cinca (2010) underscore the importance of the *perception* of risk, in this case about how perceptions of liquidity conditions affect investors' incentives and decision-making. Therefore, in addition to providing 'emergency relief' via liquidity injections when crises arise, reducing investors' perceived probability of liquidity shocks, π, merits policy consideration.

Effective policy provisions to reduce deviations from UIP along the lines mentioned above will promote efficiency in the international allocation of resources. They will reduce the cost of capital by reducing the liquidity premium which arises from potential liquidation costs. Examining the role of liquidity constraints in the failure of UIP helps us

Table 2
Slope coefficient of the forward premium, $(1+\beta)$, in $\ell-$ UIP model†††

Asset price change ($-k$)	Liq. Shock (ℓ)	$\pi = 1\%$							$\pi = 2\%$						
		$\lambda = 0$	$\lambda = 2$			$\lambda = 5$			$\lambda = 0$	$\lambda = 2$			$\lambda = 5$		
			$r = 3\%$	5%	10%	$r = 3\%$	5%	10%		$r = 3\%$	5%	10%	$r = 3\%$	5%	10%
-5%	0.88	0.99	0.99	0.99	0.98	0.99	0.98	0.97	0.98	0.98	0.97	0.97	0.97	0.96	0.94
-10%	1.81	0.98	0.97	0.97	0.96	0.96	0.95	0.89	0.96	0.95	0.94	0.92	0.93	0.90	0.79
-15%	2.79	0.97	0.96	0.95	0.92	0.94	0.90	0.71	0.93	0.91	0.90	0.84	0.87	0.80	0.41
-20%	3.83	0.95	0.94	0.92	0.86	0.89	0.81	0.25	0.91	0.87	0.84	0.72	0.78	0.62	-0.51
-25%	4.94	0.94	0.91	0.88	0.76	0.83	0.67	-0.87	0.88	0.82	0.76	0.52	0.66	0.33	-2.78
-30%	6.12	0.93	0.88	0.83	0.61	0.74	0.41	-3.71	0.85	0.75	0.66	0.21	0.48	-0.19	
-35%	7.39	0.91	0.84	0.76	0.36	0.61	-0.04	-11.2	0.82	0.68	0.52	-0.29	0.21	-1.10	
-40%	8.77	0.89	0.79	0.67	-0.05	0.41	-0.86		0.79	0.57	0.33	-1.12	-0.19		
-45%	10.26	0.88	0.73	0.53	-0.75	0.10	-2.40		0.75	0.45	0.06		-0.83		
-50%	11.90	0.86	0.64	0.34	-2.00	-0.40	-5.43		0.71	0.28	-0.33				
-55%	13.70	0.83	0.53	0.06	-4.33	-1.24			0.67	0.05	-0.90				
-60%	15.73	0.81	0.38	-0.37		-2.69			0.62	-0.26					
-65%	18.02	0.78	0.16	-1.07					0.56						
-70%	20.66	0.75	-0.16	-2.25					0.50						
-75%	23.79	0.71	-0.68						0.42						
-80%	27.62	0.67							0.33						
-85%	32.56	0.61							0.21						
-90%	39.52	0.52							0.04						
-95%	51.41	0.38													

††† Notes.

- π is the probability of a liquidity shock, ℓ is the size of the liquidity shock, r is the interest rate, and λ is the risk aversion. ℓ is obtained from (24) for the asset price change $(-k)$ in the first column.
- The forward premium bias or deviation from UIP equals (*Slope coefficient* $-1) = \beta$. From (21) $\beta = -\pi\ell(1-\pi)^{-1}\ \overline{\theta}(r,r^*)$. The forward premium $x = (r - r^*)$ and liquidity premium βx are negatively correlated since $\beta < 0$. This property is discussed in section 4.2.
- If there is no liquidity risk then $\pi = 0$. In this case $\beta = 0$, the slope coefficient is 1, and the liquidity shock channel of UIP failure ceases to exist. Under risk neutrality $\lambda = 0$, $\overline{\theta} = 1$ and $\beta = -\pi\,\ell\,(1-\pi)^{-1}$. In this case the slope coefficient is less than one but positive: $1 > (1+\beta) > 0$. Under risk aversion $\lambda > 0$, $\overline{\theta} > 1$ and the slope coefficient can be negative, as shown in cells below dashed lines. Blank cells indicate that sufficiency conditions in Appendix A are not satisfied and that the investor is better off not participating in the market.

understand the context of these policy challenges, and they remind us why it is important to obtain better insights into the forward premium puzzle.

6. Concluding remarks

In this paper I examine the role of liquidity factors in explaining the empirical failure of UIP and resolving the forward premium puzzle. In the normal state there is no liquidation of assets, but in a liquidity crisis some investors are forced to immediately liquidate their assets. In liquidity-constrained markets, these liquidations cause sudden asset price declines, subjecting investors to losses. Thus, the incentives for international interest arbitrage are affected by an anticipated liquidity shock. If the high interest rate asset brings a larger loss in a crisis, then *ceteris paribus* investors find it less desirable to hold it. When they place a high utility value on the loss, then they will hold the low interest asset even if its underlying currency is expected to depreciate.

Bansal (1997, 370) has noted that "The forward premium puzzle poses a formidable challenge since a wide array of economic models cannot explain [the] empirical finding" that the slope coefficient in UIP regressions is less than one or negative. Liquidity shocks offer a new channel for explaining persistent deviations from UIP – the forward premium puzzle. I show that the forward premium cannot be a rational forecast of the currency depreciation because forecast errors contain systematic variations in the liquidity premium even under risk neutrality. In this liquidity risk-augmented formulation of UIP, or $\ell-$ UIP, investors' required liquidity premium places a wedge between the expected currency depreciation and the interest rate differential. This liquidity premium shows up in the data as persistent excess returns. The failure of UIP occurs regardless of whether investors are risk-averse or risk-neutral.

Using a sample of nine major currencies for 1978:08–2019:09, I show that the estimated slope coefficients of UIP regressions are in line with those in the literature, and that implied $\ell-$ UIP slope coefficients are comparable to these estimates. The implications of this analysis for policies to reduce deviations from UIP are considered, including maintaining a low interest rate regime, effective macroprudential measures for liquidity provision, and considering ways of influencing perceptions of future liquidity conditions.

This research adds to the literature on the forward premium puzzle but has limitations. For instance, stronger modelling of how $\overline{\theta}$ affects the slope coefficient can yield valuable insights, and this paper provides a basis for that effort. The assumptions in modelling a liquidity shock are notional, leaving room for refinement by incorporating crisis policy shocks and amplifications that result in liquidity spirals as discussed in Krishnamurthy (2010). Chinn (2006) has found that the forward premium puzzle is less pronounced in longer relative to shorter horizons; it will be useful to examine if this result remains true in the context of this model. It is also known from Bansal and Dahlquist (2000), Frankel and Poonawala (2010), and Lee (2011) that developed and developing countries exhibit cross-sectional differences with respect to empirical violation of UIP. Likewise, in a sub-sample of the ASEAN grouping, Tang (2011) finds that UIP holds for high-income Singapore, but not for the middle-income countries in the sample. Such issues bear consideration in the present framework and are subjects for further inquiry.

Declaration of competing interest

None.

Appendix A. Derivation of the conditions for $E(u_t^H) > u(0)$ to hold

Since $u(0) = 0$, $E(u_t^H) > 0$ by second-order linear approximation of the RHS of (10) around 0 if:

$$(1-\pi)[u'0)r_t + \frac{1}{2}u''0)r_t^2] + \pi[u'0) - \ell\, r_t) + \frac{1}{2}u''0)(-\ell\, r_t)^2] > 0 \tag{A1}$$

Dividing (A1) by $(1-\pi)$ and collecting terms yields:

$$\left[1 - \frac{\pi\,\ell}{1-\pi}\right]u'(0)r_t + \left[1 + \frac{\pi(-\ell)^2}{1-\pi}\right]\frac{1}{2}u''(0)r_t^2 > 0 \tag{A2}$$

Since the absolute risk aversion parameter is $\lambda = -\frac{1}{2}\frac{u''(0)}{u'(0)}$, $E(u_t^H) > 0$ only if:

$$\left[1 - \frac{\pi\,\ell}{1-\pi}\right] > \left[1 + \frac{\pi(-\ell)^2}{1-\pi}\right]\lambda r_t > 0 \tag{A3}$$

For $E(u_t^H) > u(0)$the necessary and sufficient condition under risk neutrality $(\lambda = 0)$ is:

$$1 - \frac{\pi\,\ell}{1-\pi} > 0 \tag{A4}$$

or that

$$\pi < \frac{1}{1+\ell} \tag{A5}$$

(A5) is necessary but not sufficient under risk aversion; the first inequality in (A3) requires additional parameter restrictions, otherwise in the measure of the liquidity pressure $\ell = \delta\varphi$, δ and φ could not both be positive as theory requires. The necessary and sufficient condition under risk aversion is:

$$r_t < \frac{1-\pi(1+\ell)}{\lambda\,[1-\pi(1-\ell^2)]} \tag{A6}$$

In the text (A5) & (A6) are (32) and (33), respectively. Similar conditions apply to the Foreign investor except that the middle term of (A3) includes r_t^* instead of r_t.

Appendix B. Derivation of Equation (18)

Using equations (10)-(11) yields the following equilibrium condition for the Home investor:

$$(1-\pi)\left[u(r_t) - u\left(r_t^* + E(\tilde{d}_{t+1}) - \mu_t\right)\right] + \pi\left[u(-\ell\, r_t) - u\left(-\ell\, r_t^*\right) = 0\right. \tag{B1}$$

Similarly, equations (16)-(17) yield the equilibrium condition for the Foreign investor:

$$(1-\pi)\left[u\left(r_t^*\right) - u(r_t - E(\tilde{d}_{t+1}) - \mu_t)\right] + \pi\left[u\left(-\ell\, r_t^*\right) - u(-\ell\, r_t)\right] = 0 \tag{B2}$$

Now we express (B1) and (B2) to linear approximations around r_t and r_t^* to yield the respective zero-arbitrage conditions for Home and Foreign investors.

With regard to the Home investor, in (B1) the first-order expansions of $u(r_t^* + E(\tilde{d}_{t+1}) - \mu_t)$ aroundr_t, and then $u(-\ell\, r_t^*)$ around $-\ \ell\, r_t$,respectively yield:

$$u\left(r_t^* + E(\tilde{d}_{t+1}) - \mu_t\right) = u(r_t) + u'(r_t)\left[u\left(r_t^* - r_t + E(\tilde{d}_{t+1}) - \mu_t\right)\right] \tag{B3}$$

$$u\left(-\ell\, r_t^*\right) = u(-\ell\, r_t) + u'(-\ell\, r_t)\left[-\ell\,\left(r_t^* - r_t\right)\right] \tag{B4}$$

Substituting the RHS of (B3) and (B4) in (B1) upon rearrangement yields:

$$(1-\pi)u'(r_t)[E(\tilde{d}_{t+1}) - \mu_t] + [(1-\pi)u'(r_t) - \pi\,\ell u'(-\ell\, r_t)]\left(r_t^* - r_t\right) = 0 \tag{B5}$$

Upon dividing (B5) by $(1-\pi)u'(r_t)$ and rearranging we obtain:

$$E\left(\tilde{d}_{t+1}\right) - \mu_t = \left[1 - \frac{\pi\ell}{1-\pi}\left(\frac{u'(-\ell\, r_t)}{u'(r_t)}\right)\right]\left(r_t - r_t^*\right) \tag{B6}$$

The Home investor's zero arbitrage condition in (B6) can now be written as:

$$E(\tilde{d}_{t+1}) - \mu_t = \left[1 - \frac{\pi \ell}{1-\pi}\theta(r_t)\right](r_t - r_t^*) \tag{B7}$$

where

$$\theta(r_t) = \frac{u'(-\ell r_t)}{u'(r_t)}$$

With regard to the Foreign investor, in (B2) linearize the term $u(r_t - E(\tilde{d}_{t+1}) - \mu_t)$ around r_t^*, and then $u(-\ell r_t)$ around $u(-\ell r_t^*)$. Then proceed as above to obtain the Foreign investor's zero arbitrage condition:

$$E(\tilde{d}_{t+1}) + \mu_t = \left[1 - \frac{\pi \ell}{1-\pi}\theta(r_t^*)\right](r_t - r_t^*) \tag{B8}$$

By eliminating μ_t from the zero-arbitrage conditions in (B7) and (B8) we obtain the following international financial market equilibrium condition:

$$E(\tilde{d}_{t+1}) = \left[1 - \frac{\pi \ell}{1-\pi}\overline{\theta}(r_t, r_t^*)\right](r_t - r_t^*) \tag{B9}$$

where

$$\overline{\theta} = \frac{1}{2}\left[\theta(r_t) + \theta(r_t^*)\right] \tag{B10}$$

Equation (B9) is equation (18) in the text.

Appendix C. Derivation of $\theta'(\cdot)$

For $y_t = (r_t, r_t^*)$ equation (19) implies:

$$\theta'\left(y_t\right) = -\ell\left(\frac{u''(-\ell y_t)}{u'(y_t)}\right) - \left(\frac{u''(y_t)}{u'(y_t)}\right)\theta\left(y_t\right) \tag{C1}$$

Substituting in (C1) the expression for $\theta(y_t)$ from (19) yields:

$$\theta'\left(y_t\right) = \ell\left(-\frac{u''(-\ell y_t)}{u'(-\ell y_t)}\frac{u'(-\ell y_t)}{u'(y_t)}\right) + \left(-\frac{u''(y_t)}{u'(y_t)}\frac{u'(-\ell y_t)}{u'(y_t)}\right) \tag{C2}$$

The risk aversion parameter is:

$$\lambda = -\frac{u''(-\ell y_t)}{u'(-\ell y_t)} = -\frac{u''(y_t)}{u'(y_t)} \tag{C3}$$

Substituting the RHS of (C3) in (C2) yields:

$$\theta'(y_t) = \lambda(1+\ell)\theta(y_t) \tag{C4}$$

Data Appendix

Monthly spot rates are from *International Financial Statistics* (International Monetary Fund, 2019).

Legacy rate of DEM/EUR starting 1999:01 obtained from European Central Bank: http://www.ecb.int/press/pr/date/1998/html/pr981231_2.en.html.

Interest rates are monthly Eurocurrency rates from (Datastream, 2019). The original data are presented as annualized interest rates per cent (r_{Annual}) and were converted into monthly rates ($r_{Monthly}$) using $r_{Monthly} = (1 + 0.01 r_{Annual})^{1/12} - 1$.

Descriptive Statistics

Spot Rates	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	Obs.
Belgian Franc (BEF)	0.0286	0.0289	0.0391	0.0150	0.0049	−0.5836	2.9399	494
Canadian Dollar (CAD)	0.8057	0.8012	1.0468	0.6249	0.0980	0.4203	2.5794	494
Swiss Franc (CHF)	0.7626	0.7231	1.2807	0.3556	0.1977	0.2281	2.1352	494
Deutschemark (DEM)	0.5769	0.5847	0.8063	0.3022	0.1017	−0.4998	2.8728	494
French Franc (FFR)	0.1786	0.1779	0.2476	0.0989	0.0300	−0.1603	2.9246	494
British Pound (GBP)	1.6405	1.6042	2.4158	1.0957	0.2370	0.7927	3.9157	494
Italian Lira (ITL)	0.0007	0.0007	0.0012	0.0004	0.0002	1.6861	6.3269	494
Japanese Yen (JPY)	0.0081	0.0084	0.0130	0.0037	0.0022	−0.2919	2.6743	494
Netherlands Guilder (NEG)	0.5136	0.5212	0.7156	0.2673	0.0896	−0.5231	2.9423	494

(*continued on next column*)

(*continued*)

Spot Rates	Mean	Median	Max	Min	Std. Dev.	Skewness	Kurtosis	Obs.
Interest Rates								
Belgian Franc (BEF)	0.0039	0.0029	0.0297	−0.0004	0.0038	1.5856	8.0806	494
Canadian Dollar (CAD)	0.0045	0.0038	0.0167	0.0002	0.0035	0.8236	2.9790	494
Swiss Franc (CHF)	0.0021	0.0015	0.0085	−0.0010	0.0023	0.8117	2.8094	494
Deutschemark (DEM)	0.0031	0.0029	0.0098	−0.0004	0.0025	0.4560	2.5575	494
French Franc (FFR)	0.0045	0.0031	0.0269	−0.0004	0.0042	1.3684	6.3406	494
British Pound (GBP)	0.0052	0.0047	0.0141	0.0002	0.0038	0.3385	2.0767	494
Italian Lira (ITL)	0.0057	0.0039	0.0253	−0.0004	0.0053	0.7971	3.1666	494
Japanese Yen (JPY)	0.0020	0.0004	0.0109	−0.0005	0.0025	1.0246	3.0175	494
Netherlands Guilder (NEG)	0.0033	0.0031	0.0115	−0.0004	0.0026	0.4156	2.4140	494
United States Dollar (USD)	0.0041	0.0041	0.0172	0.0001	0.0033	0.9496	3.8897	494

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